

Clade-level NNI expansion of a CCD

Motivation. The CCD0 construction *expands the split set*: for every clade partition $P \rightarrow \{L, R\}$ that could exist because its parent P and both children L, R are already clades in $\mathcal{C}$ (the set of clades with nonzero marginal support), it adds that partition even if no sampled tree realised it. Here we describe the analogous operation one level up (expanding the *clade set* $\mathcal{C}$ itself) by reading nearest-neighbour-interchange (NNI) rearrangements directly off the stored splits, without ever building a tree.

NNI at the clade level. An internal edge of a tree is a stored split $P \rightarrow \{L, R\}$: the parent clade P branches into disjoint children L, R with $L \cup R = P$. An NNI at that edge needs the grandchildren. Descend into one child, say L , with its split $L \rightarrow \{A, B\}$, and let the sibling be R . The three subtrees meeting at the edge are A, B, R . The two NNI moves recombine them so that the child of P is no longer $L = A \cup B$ but

$$A \cup R \quad \text{or} \quad B \cup R. \quad (1)$$

Because $A \subseteq L$ and $L \cap R = \emptyset$, each union in (1) is a disjoint union of size ≥ 2 and a *proper* subclade of P (it omits the nonempty B resp. A). Each novel clade carries an NNI-implied *seed split*

$$(A \cup R) \rightarrow \{A, R\},$$

whose two constituents A, R are themselves clades already in $\mathcal{C}$. The roles of L and R are symmetric, so a split $R \rightarrow \{C, D\}$ likewise yields $C \cup L$ and $D \cup L$. Every quantity (A, B, R and the parent P) is read straight from the CCD’s split table.

Algorithm. For each clade P with $|P| \geq 3$, each split $\{L, R\} \in \Pi(P)$, each choice of descend-child $X \in \{L, R\}$ with sibling Y , and each grandchild split $\{A, B\} \in \Pi(X)$, emit the candidates $A \cup Y$ and $B \cup Y$; keep those not already in $\mathcal{C}$. This is a pass over pairs of adjacent splits, costing

$$O\left(\sum_P |\Pi(P)| \cdot \max_X |\Pi(X)|\right) = O(|\text{splits}| \times \bar{d}),$$

with $\bar{d}$ the average number of splits per clade, proportional to the size of the CCD, not to the number of trees it represents.

Two pairing modes. The CCD graph mixes splits from many posterior trees, so the algorithm above can combine a split $P \rightarrow \{L, R\}$ with a grandchild split $L \rightarrow \{A, B\}$ that never occurred together in any one tree. We therefore distinguish:

- **Graph-wide** ($\mathcal{N}_{\text{gw}}$): pair *any* stored split on P with *any* stored split on its child X . Cheapest and broadest; the pass described above.
- **Co-occurring** ($\mathcal{N}_{\text{co}}$): only recombine grandchild splits that actually appeared together under P in some single base tree, the true NNI neighbourhood of the posterior sample. This is obtained by walking the stored base trees and reading A, B, R off each internal edge.

By construction $\mathcal{N}_{\text{co}} \subseteq \mathcal{N}_{\text{gw}}$: the co-occurring set is the graph-wide set restricted to split pairs realised within a single tree. The graph-wide extras are exactly the clades that *no single sampled tree is one NNI move away from*; whether that is desirable broader coverage or an overreach beyond the sample’s NNI neighbourhood is a modelling choice one should state explicitly.

Conservativeness. NNI is the most local rearrangement, so the expansion recovers only clades “near” the support and misses anything requiring an SPR-scale move. For the single tree $((A, B), C), D$ the expansion yields exactly $\{ABD, CD, AC, BC\}$; it does *not* produce AD, BD, ACD or BCD , which need a non-local move. If the concern is held-out trees that differ from the sample by a single local rearrangement, this coverage is exactly right; for broader topological novelty one would extend to SPR neighbourhoods, also expressible at the clade level (a regrafted subtree clade unioned with whatever lies below the target edge), at the cost of many more candidates.

Parameterisation in CCD1 / regCCD

The clade expansion is most natural in the *split*-parameterised models. A regularised CCD (regCCD) takes the CCD1 graph, adds the unobserved splits whose parent and children are already clades (the CCD0 split expansion), and prices every split of a clade C by additive smoothing,

$$\theta_\alpha(S) = \frac{f(S) + \alpha}{f(C) + \alpha |S(C)|}, \quad (2)$$

a Dirichlet($\alpha, \dots, \alpha$) prior over the splits $S(C)$. This gives nonzero probability to trees with *unsampled splits of sampled clades*, but a tree containing an *unsampled clade* still has probability zero: the residual limitation of both CCD0 and CCD1.

NNI clade expansion closes this gap, and, crucially, needs *no new parameter*. Under the CCD1 form $P(T) = \prod_v \text{CCP}(\text{split} \mid \text{clade})$ a clade carries no marginal parameter; its probability is induced by the product along the path to it. So adding a novel clade $K = A \cup R$ only requires its *splits* to be in the graph:

- the recombined parent split $P \rightarrow \{K, B\}$ enlarges $S(P)$ and is priced by (2) with $f = 0$, i.e. mass $\alpha/(f(P) + \alpha |S(P)|)$;
- the seed split $K \rightarrow \{A, R\}$ (and any further splits of K into existing clades found by the split expansion) is priced the same way; a clade left with a single split is deterministic ($\text{CCP} = 1$).

A held-out tree that is one NNI move from the support is then scored $P(T) = [\prod \text{observed CCPs}] \cdot \theta_\alpha(P \rightarrow \{K, B\}) \cdot \theta(K \rightarrow \{A, R\}) > 0$. The novel clades enter the *same* additive- α Dirichlet-multinomial as the observed and split-expanded splits (single pseudocount), so the only model change is a larger clade set.

Coverage vs. sharpness, and a two-level prior. The benefit is out-of-sample coverage of trees with unsampled clades; the cost is that enlarging $|S(P)|$ dilutes the smoothed mass of observed splits. The latter is controlled by giving NNI-derived splits (those touching a novel clade) a *separate* pseudocount β , while observed and split-expanded splits keep α :

$$\theta(S) = \frac{f(S) + \alpha [S \text{ regular}] + \beta [S \text{ NNI}]}{f(C) + \alpha |R(C)| + \beta |N(C)|}, \quad (3)$$

an asymmetric Dirichlet placing a weaker prior on the speculative recombinations. Crucially β does not affect *coverage* (any $\beta > 0$ keeps the NNI split positive); it only rebalances mass at observed parents.

Choosing the pseudocounts. Since β does not change which trees are covered, the held-out log probability over the (fixed) covered set is a clean objective for selecting (α, β) . Because the pseudocounts do not change the graph, the model is fit once and the (α, β) grid is scored in closed form. The optimal β is typically an order of magnitude below α , confirming the weaker prior on NNI splits; and the optimal α shifts under the expansion, so the pseudocounts should be re-tuned rather than inherited.

Worked example: held-out calibration on RSV2

The most direct test of whether the expansion is well-calibrated is to train on one posterior sample and score an *independent* posterior sample from the same MCMC analysis. We use two independent runs on RSV2 ($n = 129$, each with 18,001 post-burn-in trees) and ask, at the pipeline-fitted (α, β) , how much probability mass the model places on NNI-expanded trees versus the fraction of held-out trees that actually fall in that zone.

The full pipeline. The clade set under regCCD is built in four stages,

$$\mathcal{C}^{(1)} \xrightarrow{\text{split exp.}} \mathcal{C}^{(1)} \cdot se \xrightarrow{\text{NNI exp.}} \mathcal{C}^{(1)} \cdot se \cdot ne \xrightarrow{\text{split exp.}} \mathcal{C}^{(1)} \cdot se \cdot ne \cdot se, \quad (4)$$

each enlarging the supported tree space. Crucially, the *first* split expansion runs before the NNI pass, so the NNI recombinations get to use the CCD0-expanded split contexts; this matters for graph-wide pairing, which then finds additional novel clades that the observed-only graph would miss.

Tree-space blowup. On RSV2 the four spaces have the following sizes. Both modes share $|\mathcal{C}^{(1)}| = |\mathcal{C}^{(1)} \cdot se| = 1,919$; graph-wide adds 42,627 novel clades, co-occurring adds 7,956.

stage	clades	splits	$\log_{10} N_{\text{trees}}$
$\mathcal{C}^{(1)}$	1,919	7,653	46.33
$\mathcal{C}^{(1)} \cdot se$	1,919	23,597	50.44
$\mathcal{C}^{(1)} \cdot se \cdot ne$ (gw)	44,546	157,602	71.79
$\mathcal{C}^{(1)} \cdot se \cdot ne \cdot se$ (gw)	44,546	3,142,692	87.06
$\mathcal{C}^{(1)} \cdot se \cdot ne$ (co)	9,875	41,517	67.30
$\mathcal{C}^{(1)} \cdot se \cdot ne \cdot se$ (co)	9,875	256,497	77.92

The first split expansion contributes +4.1 decades to the tree count, the NNI clade expansion contributes +21.4 (gw) or +16.9 (co), and the final split expansion contributes a further +15.3 (gw) or +10.6 (co). The NNI clade expansion is the single largest jump in tree count, while the final split expansion contributes the largest jump in raw split count (an additional $\sim 3 \times 10^6$ partitions under graph-wide), most of which are combinations of novel clades that would never appear in a sampled tree.

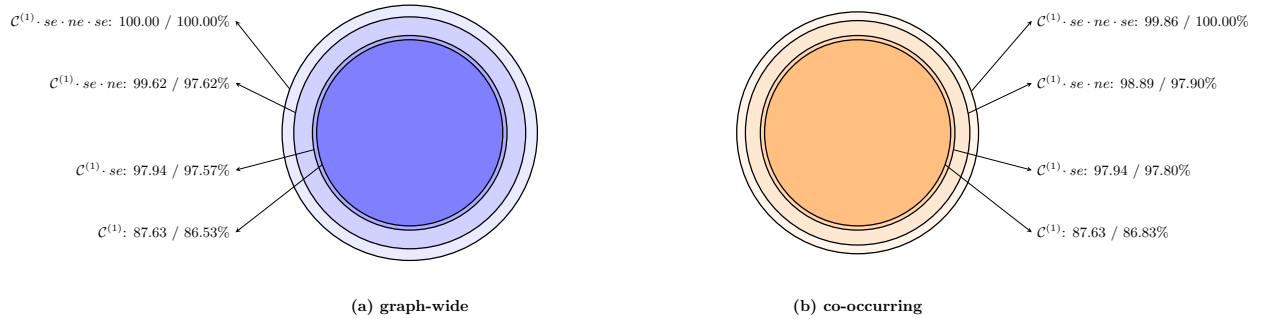


Figure 1: Tree-space envelope at each pipeline stage on RSV2 (train: run 1, 18,001 trees; held out: run 2, same size). Areas are proportional to $\log_{10} N_{\text{trees}}$ (so radii are proportional to $\sqrt{\log_{10} N}$). Each circle is annotated with the fraction of held-out trees that fall inside it (cumulative empirical coverage) and the model’s cumulative probability mass on trees inside that universe under the fitted (α, β) . The two quantities agree to within ~ 2 percentage points at every boundary in both panels, despite each transition enlarging the tree count by many decimal orders of magnitude.

Held-out fit. We train the full pipeline on run 1 and grid-search (α, β) to maximise the mean held-out log-probability on run 2 (no cross-validation needed: the two runs are genuinely independent). The optima are

mode	α^*	β^*	mean $\log P/\text{tree}$	coverage
graph-wide	0.265	0.0014	-54.21	18001/18001 (100.0%)
co-occurring	0.260	0.0084	-54.11	17976/18001 (99.86%)

The graph-wide expansion covers every held-out tree; the 25 co-occurring misses are trees whose novel clades sit more than one NNI move away from any clade-context in run 1, an SPR-scale rearrangement away.

Probability mass on NNI-expanded trees vs. empirical fraction. At the fitted (α, β) , we compute in closed form the total probability mass the model places on trees that contain at least one novel clade,

$$P(\text{novel-clade tree}) = 1 - M(\rho), \quad M(c) = \sum_{(L,R) \in S^{\text{obs}}(c)} \theta(L, R) M(L) M(R), \quad (5)$$

restricted at every clade c to the sub-sum over existing-only splits (novel children contribute $M = 0$). We then compare to the *empirical* fraction of held-out trees that actually contain a novel clade:

mode	model $P(\text{novel})$	empirical	ratio
graph-wide	2.43%	2.06%	1.18×
co-occurring	2.20%	1.92%	1.15×

The model overstates the speculative fraction by only $\sim 15\text{--}20\%$ in both modes. Given that the universe of *potentially* novel trees is $\sim 10^{40}$ times larger than the observed-clade tree set (cf. the table above), matching the empirical novel-clade rate within a factor of 1.2 is the operative sense in which the NNI clade expansion is *well-calibrated*: the additive- α, β smoothing under the pipeline of (4) carves a probability mass on the expansion zone that closely tracks the rate at which independent posterior samples actually visit it.